

Homework 4

Due: March 4, 2022

All homeworks are due at 11:59 PM on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit.

Problem 1

- a. Prove by induction that for all positive integers n , there exists a positive integer m such that:

$$m^2 \leq n < (m+1)^2$$

- b. Prove by contradiction that there exists a **unique** such m .

Problem 2

The CS22 cafe has a complex cash register system constituting of boxes that each contain pennies.

Consider 2^n pennies with $n \geq 1$ that are distributed across $m \geq 1$ boxes. We know nothing about the distribution of the pennies (some boxes could have no pennies). We are allowed to move pennies in the following way:

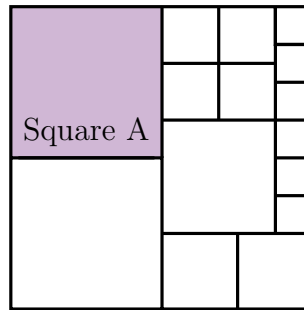
- i. Select two boxes B_i and B_j . Let B_i have P_i pennies and let B_j have P_j pennies where $P_i \geq P_j$.
- ii. You may then move P_j pennies from B_i to B_j .

In other word, you may exactly double the number of pennies in the box with fewer pennies by moving pennies from the box with more pennies.

- a. Prove via contradiction that the number of boxes containing an odd number of pennies must be even.
- b. Describe a strategy to get an even number of pennies in all boxes.
- c. Prove via induction that it is possible to get all of the pennies into one box given 2^n pennies where $n \geq 1$.

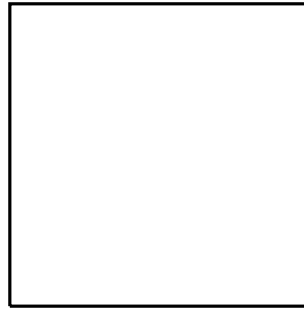
Problem 3

- a. Consider the figure below.



How does the number of (non-overlapping) squares in the figure change if Square A (highlighted) is subdivided into squares whose side lengths are half that of the original's? Note that the figure initially has 15 non-overlapping squares.

- b. Divide the square below into exactly 6 (not necessarily congruent) squares such that there is no overlap and the entire region is covered.



- c. Prove by strong induction that every integer greater than 5 can be written in the form $a + 3m$, where a is 6, 7, or 8, and m is a non-negative integer.
- d. Using parts (a), (b), and (c), prove by strong induction that a square can be subdivided into n (not necessarily congruent) squares such that there is no overlap and the entire region is covered for all integral n greater than 5.

 Problem 4 (Mind Bender — *Extra Credit*)

Mind Benders are extra credit problems intended to be more challenging than usual homework problems and are an exploration into a topic not covered in lecture. This week, we'll think about induction over sets other than $\mathbb{N}$.

First of all: what is $\mathbb{N}$, exactly? Our definition $\{0, 1, 2, \dots\}$ wasn't very precise. Thinking more carefully, you might come up with the following definition:

1. 0 is a natural number.
2. For any natural number n , **successor**(n) is a natural number.
3. The **successor** function is injective.
4. For every n , **successor**(n) $\neq$ 0.
5. Every natural number is either 0 or the successor of some other natural number. (In other words, rules 1 and 2 are the only ways to “create” natural numbers.)

We then define $\mathbb{N}$ to be the set of all natural numbers.

- a. Using this characterization of $\mathbb{N}$, explain why the principle of induction works to prove facts about all natural numbers. (Note: $n + 1$ is another way to write **successor**(n).)
- b. We need each of properties 3-5 to make sure that $\mathbb{N}$ has the shape we want.

Example

Suppose we defined $\mathcal{N} = \{0, 1, 2\}$ with

$$\text{successor}(0) = 1$$

$$\text{successor}(1) = 2$$

$$\text{successor}(2) = 0$$

$\mathcal{N}$ satisfies properties 1, 2, 3, and 5, but fails 4.

Give an example like this of a set and successor function that satisfies every property *except* property 3, and another example that satisfies every property *except* 5.

We can give a similar characterization of *lists* of elements of some set T .

1. The empty list $[]$ is a list of elements of T .
2. For any $t \in T$ and list L of elements of T , **append**(t, L) is a list of elements of T : that is, it's the list we get when we *append* t at the beginning of L .
- 3-5. Similar to 3-5 above for $\mathbb{N}$.

This may not be a definition of lists that you've seen before. We call it, maybe tellingly, an *inductive* (or *algebraic*) definition.

- c. "Translate" properties 3-5 from the $\mathbb{N}$ case so that they make sense in the list case.
- d. Lists can be a kind of mathematical object just like numbers: we may want to prove a theorem about all lists of elements of some set T . Analogous to the principle of induction for $\mathbb{N}$, there's a principle of induction for lists. State this principle of induction.
- e. We say a *subsequence* is a non-necessarily contiguous subset of a list L . For example, if L were the list $[1, 2, 3, 4, 5]$, then one such subsequence could be $[2, 4, 5]$. Prove, by inducting on the structure of lists, that the number of subsequences of a list of length n (containing all distinct elements) is 2^n .
- f. Can you think of any other sets/structures that can be defined "inductively" like this?



Note: these sets show up all the time in certain areas of computer science. Some programming languages *only* allow you to define new datatypes as inductive sets¹ (or "inductive types")! What if you write a complicated program that takes inputs from one of these datatypes, and want to prove that your program has certain behavior? You guessed it: time for induction.

¹For those who have taken a computer science course that has introduced functional programming, **append** may look very similar to **link** or **cons**!